Maxmilian Duarte

Python Backend Engineer

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SUMMARY

Backend engineer with 4 years of production Python: REST APIs, asynchronous processing with Celery and RabbitMQ, data workflows built on advanced SQL, and automation that takes manual work off operations teams. I own systems end to end: design, implementation, tests and deployment with Docker on Linux.

SKILLS

Backend	Python, Flask, FastAPI, Django, REST APIs, Celery, RabbitMQ, OAuth2 / PKCE, Server-Sent Events
Data	MySQL, SQLite, MongoDB, advanced SQL and stored procedures, ETL
Infra	Docker / Compose, Linux, systemd, nginx, Proxmox / LXC, GitHub Actions, Uptime Kuma, scheduled backups
Quality	pytest, end-to-end tests, type hints, production support (L3), log analysis
Also	TypeScript / Next.js, Vue 3, Kotlin / Jetpack Compose, Selenium / Playwright, FFmpeg

EXPERIENCE

Independent Developer
Jul 2026 – present · Remote

Own products and open source

- Building MaxTools Radio (see below) and cnabkit, an open-source Python library for Brazilian bank files (CNAB 240/400, OFX), on infrastructure I run myself: Proxmox, Docker, nginx, Uptime Kuma, weekly backups.

Netbiis | Data Driven Technologies
Sep 2024 – May 2026 · Remote

Software Engineer (Python, Backend & Data)

- Built and owned an internal Flask + SQL web application that automated data validation for campaign operations, cutting the team's manual verification time by 40%.
- Wrote and optimized complex SQL and stored procedures for high-volume extraction and audience segmentation in Adobe Campaign.
- Provided L3 support for production automation pipelines: log-based root-cause analysis, fixes and follow-up.

Independent Consultant
Feb 2024 – Aug 2024 · Hybrid

Backend Developer for HG7 and ExportSul (insurance brokerages)

The HG7 engagement began part-time in 2023, alongside the internship.

- HG7: built an OFX bank-statement import and reconciliation system (Python, Flask, MySQL) across three iterations, embedded in the company's PHP ERP and feeding its income statement (DRE), replacing line-by-line spreadsheet reconciliation.
- HG7: integrated RD Station CRM and 3CX telephony with internal systems; ETL from RD Station and Microsoft Graph into MongoDB for Power BI dashboards.
- ExportSul: Selenium bots processing invoices in an ERP, with retries and per-item state so runs resume where they stopped.

Plug Benefícios
Aug 2022 – Feb 2024 · São Paulo

Python Developer (Internship)

- Built and maintained 15 production bots (Python, Selenium, Playwright) processing beneficiary changes across 4 health-insurer portals from a MySQL ticket queue, plus invoices, slips and payslips.
- Designed a Flask + Celery + RabbitMQ service that runs the bots in parallel behind a REST API, with concurrency sized by a load test.
- Extracted shared automation code into an internal Python package (wheel) reused by every bot; MySQL for transactional data, MongoDB for scraped data, internal dashboards in Django and Flask.

Microcamp Tecnologia
Oct 2021 – Feb 2022 · São Paulo

Tech Instructor: programming logic, algorithms and Python for 50+ students

SELECTED PROJECTS — case studies at maxmilian02.github.io

MaxTools Radio — Python bot streaming 48 kHz PCM through a real-time DSP chain, Flask API with SSE and OAuth accounts (Google, Discord, Spotify), web app and Kotlin / Jetpack Compose Android app. 1,180 Python tests, 340 Android unit tests.

MaxTools — FastAPI plugin platform: a registry auto-discovers tool modules and a Next.js dashboard renders from their manifest. Async jobs with progress, API-key auth, TTL cleanup and frame-accurate video trimming over HTTP Range.

EDUCATION · LANGUAGES

B.Sc. Computer Engineering, Impacta Tecnologia (expected 2027)
Portuguese (native) · English (advanced: reading, writing, technical documentation)